JIANAN MU



Assistant Researcher
State Key Laboratory of Processors
Institute of Computing Technology, Chinese Academy of Sciences | Beijing, China
mujianan@ict.ac.cn | Scholar Page: scholar.google.com/citations?user=_urdIzIAAAAJ

RESEARCH PROFILE

Researcher in design verification and validation, with a focus on efficient RTL simulation, parallel fault simulation, AI-assisted verification, and functional safety. He has authored or co-authored more than 20 papers as first or corresponding author in top-tier conferences and journals in EDA and AI, including DAC, ICCAD, TCAD, MICRO, ICML, and ICLR.

EDUCATION

Ph.D., Institute of Computing Technology, Chinese Academy of Sciences, 2024.
Bachelor's degree, School of Electronics Engineering and Computer Science, Peking University, 2019.

SELECTED DISTINCTIONS

Outstanding Doctoral Dissertation Award (2025), China Computer Federation, Technical Committee on Integrated Circuit Design (CCF TCICD).

ICCAD 2024 — Best Paper Award Nomination, DDP-fsim. Corresponding author.

ASP-DAC 2025 — Best Paper Award Nomination, ETPG. Corresponding author.

SELECTED PAPERS: SIMULATION AND FUNCTIONAL SAFETY

ARMOR: Accelerating RTL Simulation by Mitigating the Front-End Bottleneck Using Node Compression
J. Tang, **J. Mu**, Z. Chao, J. Wen, J. Ye, and H. Li.
MICRO 2026, accepted | Corresponding author

DDP-fsim: Efficient and Scalable Fault Simulation for Deterministic Patterns with Two-Dimensional Parallelism
F. Gu, M. Wang, **J. Mu**, Z. Liu, J. Tang, H. Wang, Y. Wang, J. Ye, H. Li, and X. Li.
ICCAD 2024 | Best Paper Award Nomination; corresponding author

ETPG: Efficient Transition Fault Simulation via Dual-Strategy Pattern Parallelism and Gate Restructuring
M. Wang, H. Wang, Z. Liu, F. Gu, **J. Mu**, J. Tang, J. Gao, H. Li, J. Ye, and X. Li.
ASP-DAC 2025 | Best Paper Award Nomination; corresponding author

RIROS: A Parallel RTL Fault SImulation FRamework with TwO-Dimensional Parallelism and Unified Schedule
J. Tang, **J. Mu**, Z. Liu, G. Yu, T. Hua, B. Sun, S. Liu, J. Ye, and H. Li.
ICCAD 2025 | Corresponding author

EPICS: Efficient Parallel Pattern Fault Simulation for Sequential Circuits via Strongly Connected Components
M. Wang, H. Wang, **J. Mu**, X. Zhang, B. Sun, Y. Wen, et al.
DAC 2025 | Corresponding author

PastATPG: A Hybrid ATPG Framework for Better Test Compaction with Partial Assignment SAT
Z. Chao, X. Zhang, X. Zhang, **J. Mu**, Z. Liu, S. Liang, S. Cai, J. Ye, X. Li, and H. Li.
DAC 2025 | Corresponding author

DomSim: Hardware-Aware Hybrid Fault Simulation with Dominator Tree-guided Partitioning
M. Wang, H. Wang, F. Gu, Z. Liu, **J. Mu**, et al.
IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, 2025 | Corresponding author

Bit-Compressed Concurrent Fault Simulation with Efficient Fault List
F. Gu, Z. Chao, H. Wang, **J. Mu**, M. Wang, J. Gao, Y. Wang, J. Ye, X. Li, and H. Li.
IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, 2026 | Corresponding author

SELECTED PAPERS: AI-ASSISTED VERIFICATION AND RTL DESIGN

Bridging Layout and RTL: Knowledge Distillation Based Timing Prediction
M. Wang, Y. Wen, B. Sun, **J. Mu**, J. Li, X. Wang, J. J. Ye, B. Yu, and H. Li.
ICML 2025 | Spotlight; corresponding author

FicGCN: Unveiling the Homomorphic Encryption Efficiency from Irregular Graph Convolutional Networks
Z. Kan, H. Han, S. Shi, T. Hua, H. Lu, X. Li, **J. Mu**, and X. Hu.
ICML 2025 | Corresponding author

Chimera: A Unified FHE Accelerator with Enhanced Compatibility for TFHE
T. Hua, H. Li, **J. Mu**, J. Ye, L. Kong, and M. Zhang.
DAC 2026 | Corresponding author

MOSS: Multi-Modal Representation Learning on Sequential Circuits
M. Wang, B. Sun, **J. Mu**, F. Gu, B. Han, T. Yang, X. Zhang, S. Liu, Y. Wen, et al.
DAC 2025 | Corresponding author

Alchemist: A Unified Accelerator Architecture for Cross-Scheme Fully Homomorphic Encryption
J. Mu, H. Han, S. Shi, J. Ye, Z. Liu, S. Liang, M. Li, M. Zhang, S. Bian, X. Hu, H. Li, et al.
DAC 2024 | First author

Think with Self-Decoupling and Self-Verification: Automated RTL Design with Backtrack-ToT
Z. Chao, Y. Wang, X. Zhang, J. Zhou, T. Hua, H. Han, T. Yang, **J. Mu**, et al
DATE 2026 | Corresponding author

Scalable and Conflict-Free NTT Hardware Accelerator Design: Methodology, Proof, and Implementation
J. Mu, Y. Ren, W. Wang, Y. Hu, S. Chen, C. H. Chang, J. Fan, J. Ye, Y. Cao, H. Li, et al.
IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, 2023 | First author

TESLA: Testability Enhancement for Shift-Left Automation via Multi-LLM Collaboration
Z. Chao, R. Zhang, F. Gu, H. Lyu, B. Sun, W. Li, Z. Liu, **J. Mu**, J. Ye, X. Li, and H. Li.
ITC 2025

Faver: Boosting LLM-based RTL Generation with Function Abstracted Verifiable Middleware
J. Mu, M. Shi, Y. Wang, T. Yang, B. Sun, X. Hu, J. Ye, and H. Li.
Journal of Computer Research and Development | First author

Pecker: Bug Localization Framework for Sequential Designs via Causal Chain Reconstruction
J. Tang, **J. Mu**, T. Ma, Z. Chao, J. Ye, and H. Li.
ACM Transactions on Architecture and Code Optimization (TACO) | Under review; corresponding author

From Indiscriminate to Targeted: Efficient RTL Verification via Functionally Key Signal-Driven LLM Assertion Generation
Y. Wang, H. Lyu, B. Chen, M. Bao, W. Ding, F. Gu, Z. Chao, **J. Mu**, et al.
arXiv:2604.08932, 2026.

BROADER AI FOR EDA CONTRIBUTIONS

CircuitNet 3.0: A Multi-Modal Dataset with Task-Oriented Augmentation for AI-Driven Circuit Design
ICLR 2026 | Co-corresponding author

QiMeng-CodeV-R1: Reasoning-Enhanced Verilog Generation
NeurIPS 2025 | Co-author

PROFESSIONAL SERVICE

Journal reviewer: IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD); IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I).

Conference reviewing and PC/TPC service: AAAI Conference on Artificial Intelligence (AAAI); International Conference on Solid-State and Integrated-Circuit Technology (ICSICT); Asian Test Symposium (ATS).

Speaking and session moderation: CCF CHIP 2026; “LLM for Processor Chip Design” Summer School, Institute of Computing Technology, CAS, July 2025.